



# A two-stage fine-tuning strategy for 3D object segmentation from multi-view images

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## ABSTRACT

Neural Radiance Fields (NeRF) have been widely adopted for reconstructing high-quality 3D scenes from 2D RGB images. However, achieving accurate 3D object segmentation within these reconstructed scenes remains challenging. Existing NeRF-based segmentation methods either rely on post-processing (SA3D), which produces noisy point clouds due to the absence of density field optimization, or employ joint training with additional segmentation heads (FruitNeRF), which can lead to suboptimal performance due to conflicting learning objectives. In this work, we propose InvNeRF-Seg (Input-substitution NeRF for Segmentation), a two-stage fine-tuning strategy for 3D object segmentation that preserves the original NeRF architecture and loss function entirely. We first train a standard NeRF on RGB images and then fine-tune it using 2D segmentation masks formatted as RGB-like inputs, without introducing any architectural modifications or additional loss functions. This input-substitution approach reshapes the density field to align with object regions while suppressing background density. We validate InvNeRF-Seg through comprehensive ablation studies examining the roles of density and color MLPs, loss function choices, and training strategies. Field density analysis reveals consistent semantic refinement: densities of object regions increase while background densities are suppressed. Experiments on synthetic fruit datasets and real-world soybean imagery demonstrate that InvNeRF-Seg produces cleaner 3D segmented point clouds compared to both SA3D and FruitNeRF, enabling more accurate downstream object counting. The method is further validated on a self-collected soybean dataset to demonstrate its applicability in real-world agricultural scenarios. Our code is available at <https://github.com/ZJiangsan/InvNeRF-Seg>.

## 1. Introduction

Neural Radiance Fields (NeRF) have demonstrated remarkable success in reconstructing photo-realistic 3D scenes from 2D images by learning a volumetric scene representation [1–3]. However, while NeRF-based models have been widely adopted for view synthesis, research into object-level 3D segmentation within reconstructed NeRF scenes remains relatively limited [1,4,5].

Accurate 3D segmentation is essential for downstream applications such as robotic perception [6], augmented reality (AR) [7], autonomous driving [8], medical imaging [9], and plant phenotyping in agriculture [4,10]. However, existing approaches to 3D segmentation either rely on fully annotated point clouds—requiring substantial manual labeling [11]—or introduce significant architectural complexity by coupling additional semantic branches or loss functions into NeRF models [4,5]. Alternatively, post-processing-based methods like SA3D [1] perform 3D

segmentation by back-projecting 2D masks, but suffer from noise and lack of density refinement. These limitations hinder the adoption of NeRF-based segmentation in real-world applications where robustness and ease of implementation are critical.

A key observation motivating this work is that directly training NeRF on binary segmentation masks—which lack the smooth photometric gradients essential for geometry learning—leads to complete failure, producing entirely black, degenerate outputs (Fig. 1). However, when a NeRF model has already learned accurate geometry from RGB images, fine-tuning on binary masks can effectively reshape the density field to focus on foreground objects while suppressing background density.

Motivated by these challenges, we introduce InvNeRF-Seg, a simple yet effective two-step fine-tuning method for NeRF-based 3D object segmentation. The method maintains the original NeRF architecture and loss function entirely unchanged. The first step involves training a high-quality NeRF model on multi-view RGB images using a standard NeRF

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pipeline. In the second step, the pretrained NeRF is fine-tuned using 2D segmentation masks of the same images—without altering any architectural components—to derive a 3D segmentation-aware model.

Despite its simplicity, InvNeRF-Seg produces high-quality segmented 3D point clouds across both synthetic fruit datasets and real-world plant imagery, offering a practical bridge between 2D mask supervision and 3D semantic reconstruction. To ensure practical relevance, we further validate the proposed method on a self-collected soybean dataset acquired under real-world conditions. The main contributions of this work are summarized as follows:

- (1) We demonstrate that 3D segmentation in NeRF can be achieved through mask-based fine-tuning that reshapes the volumetric density field without requiring architectural modifications.
- (2) We provide field-level analysis revealing consistent density redistribution — foreground reinforced, background suppressed — offering insight into how supervision influences implicit neural representations.
- (3) We show through systematic ablation studies that this geometry-first, semantics-later strategy outperforms joint training, leading to cleaner 3D segmentation and more accurate downstream counting.

The rest of the paper is organized as follows: [Section II](#) reviews related work on NeRF-based 3D segmentation. [Section III](#) presents the details of the proposed InvNeRF-Seg method. [Section IV](#) provides a comprehensive evaluation including ablation studies. [Section V](#) discusses key insights and practical considerations. Finally, conclusions are drawn in [Section VI](#).

## 2. Related work

Classical deep learning methods for 3D segmentation, such as PointNet [11] and PointNet++ [12], have achieved remarkable success. However, their reliance on labor-intensive and time-consuming 3D point cloud annotation remains a significant bottleneck, limiting their scalability and applicability across domains [13]. Additionally, these methods often struggle in scenarios with noisy or incomplete point clouds, especially in complex environments with occlusions [14,15].

Neural Radiance Fields (NeRF) have transformed 3D reconstruction through learning implicit neural representations of scenes from multi-view RGB images, enabling photo-realistic novel view synthesis [2]. Building on this foundation, several approaches have been developed on top of NeRF to achieve 3D segmentation with the help of segmented 2D segmentation masks. Among these, two prominent categories have emerged: Segment Anything in 3D (SA3D) [1] and FruitNeRF [4].

### 2.1. SA3D: post-processing approach

SA3D is a post-processing method that does not change the NeRF model or training. Instead, it uses a pretrained RGB NeRF and performs 3D segmentation by projecting 2D binary masks into the scene using the rendered weights  $\omega_i$  for each sample point  $X_i$  along ray  $r$ . The weights from volume rendering are computed as:

$$\omega_i = T_i \cdot (1 - \exp(-\sigma_i \delta_i)) \cdot m_i \quad (1)$$

where  $T_i$  is the accumulated transmittance,  $\sigma_i$  is the predicted density,  $\delta_i$  is the distance between sampled points, and  $m_i$  is the binary mask value for the corresponding 2D pixel. For all rays  $r$  whose corresponding 2D pixels are labeled as foreground in the segmentation mask (i.e.,  $M(r) = 1$ ), SA3D collects 3D points where the rendering weights exceed a threshold  $\tau$ :

$$P\_SA3D = \{X_i | \sum_r M(r) \cdot \omega_i(r) \geq \tau\} \quad (2)$$

No backpropagation or model update is involved in SA3D. Even though it is fast, the noisy and fragmented segmentation point cloud affects subsequent analysis negatively because no density field optimization occurs. Although SA3D incorporates self-prompting strategies and IoU-aware view rejection to improve segmentation robustness, these techniques primarily address errors in 2D mask generation rather than 3D point cloud quality.

### 2.2. FruitNeRF: joint training approach

FruitNeRF extends NeRF by adding a segmentation branch that outputs a scalar foreground probability  $m(t) \in [0,1]$  at each sampled point along a ray. The output of the modified NeRF is:

$$f\theta(X, d) \rightarrow (c, \sigma, m) \quad (3)$$

where  $c$  is RGB color,  $\sigma$  is volume density, and  $m$  is segmentation confidence. The rendered mask is computed with the same volume rendering accumulation used for RGB color:

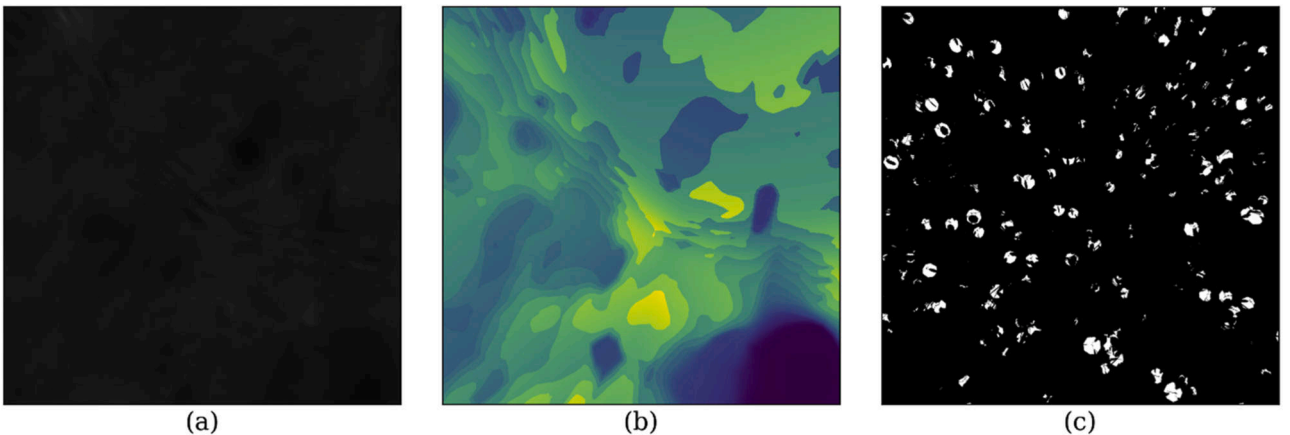
$$\widehat{W}(r) = \sum T_i (1 - \exp(-\sigma_i \delta_i)) \cdot m_i \quad (4)$$

FruitNeRF uses a composite loss:

$$\mathcal{L}_{FruitNeRF} = \mathcal{L}_{RGB} + \varphi \mathcal{L}_{seg} \quad (5)$$

where  $\mathcal{L}_{RGB}$  is the mean squared error (MSE) loss on RGB reconstruction,  $\mathcal{L}_{seg} = \text{BCE}(\widehat{W}, M_{gt})$  is the binary cross-entropy (BCE) between rendered and ground truth masks, and  $\varphi$  is the weighting factor between RGB and segmentation tasks, set to 1 by default.

FruitNeRF adopts a joint training strategy, where a single model is



**Fig. 1.** Training NeRF from scratch on binary masks. (a) Predicted output (completely black), (b) predicted depth (no structure), (c) ground truth mask. Training on masks without pretrained geometry fails completely.

trained to simultaneously predict RGB colors and segmentation masks. While FruitNeRF has demonstrated success in segmenting various fruit types, the joint training process must balance two inherently different objectives, which can lead to suboptimal performance in both RGB reconstruction and mask prediction. This problem becomes more pronounced when the objects of interest are small, since the segmentation loss contributes less to the overall optimization compared to the RGB loss. InvNeRF-Seg addresses the limitations of both SA3D and FruitNeRF by simplifying the model training process while achieving better performance in 3D segmentation.

### 3. Proposed approach

#### 3.1. Motivation

NeRF models rely heavily on smooth photometric gradients—such as color, shading, and texture—to infer scene geometry [2]. As a result, directly training NeRF using binary segmentation masks, which lack these properties, typically leads to failure. The model produces entirely black outputs with no meaningful geometry (Fig. 1), because binary masks provide insufficient gradient signals for learning a coherent density field from scratch.

However, a NeRF trained on RGB images already encodes accurate geometry. We hypothesized that fine-tuning the model on 2D binary masks—without changing the model or loss functions—could shift the density field to focus on foreground objects. This simple fine-tuning strategy proved effective, resulting in high-quality segmentation in both rendered 2D masks and 3D point clouds.

#### 3.2. Pipeline formulation

##### 3.2.1. Training a standard NeRF model based on RGB images

NeRF learns a continuous function  $f_\theta$  to map a 3D spatial coordinate  $X \in \mathbb{R}^3$  and a viewing direction  $d \in \mathbb{R}^3$  to an emitted color  $c \in \mathbb{R}^3$  and a volume density  $\sigma \in \mathbb{R}_{\geq 0}$  based on multi-view 2D RGB images with known camera poses:

$$f_\theta(X, d) \rightarrow (c, \sigma) \quad (6)$$

Points are sampled from rays marched from each pixel to camera origin within the distance between near  $a$  and far  $b$  planes and then rendered to RGB values:

$$\hat{C}(r) = \int_a^b T(t) \sigma(t) c(t) dt \quad (7)$$

where the transmittance is:

$$T(t) = \exp\left(-\int_a^t \sigma(s) ds\right) \quad (8)$$

The training MSE between rendered and ground-truth RGB values is:

$$\mathcal{L}_{RGB} = \sum_{r \in R} \|\hat{C}(r) - C_{gt}(r)\|^2 \quad (9)$$

##### 3.2.2. Fine-tune the trained standard NeRF using segmented 2D masks

The model architecture and loss functions are unchanged compared to a standard NeRF model. During fine-tuning, we replace RGB images with binary masks formatted as RGB-like inputs (i.e., [1,1,1] for object, [0,0,0] for background). The loss remains as MSE:

$$\mathcal{L}_{mask} = \sum_r \|\hat{C}(r) - M_{gt}(r)\|^2 \quad (10)$$

where  $\hat{C}(r)$  is the rendered RGB prediction (interpreted as soft mask) and  $M_{gt} \in \{0,1\}^3$  is the binary mask target. No semantic supervision or binary loss (e.g., BCE) is introduced in InvNeRF-Seg. The model is implicitly encouraged to reshape its density field to align with the masked object regions, since only those regions contribute to the loss. This formulation preserves the original NeRF architecture, training

pipeline, and loss function, but shifts the model's learning objective via input substitution alone. Because Stage 2 fine-tunes the entire network on masks formatted as RGB inputs, the resulting checkpoint is a dedicated segmentation model: its rendered output encodes foreground probability rather than scene appearance, and the original colour, shading, and texture are intentionally overwritten. The two stages are stored as separate checkpoints. The Stage-1 RGB model is retained both as the initialization for fine-tuning and for any task that still requires photometric rendering, while the Stage-2 model is used exclusively for segmentation and point-cloud extraction. This separation is by design—appearance and semantics are represented by two specialized models rather than one multi-task network—which is precisely what allows each to be optimized without compromise.

### 4. Experiments and results

#### 4.1. Dataset

We used the apple and peach datasets from synthetic fruit trees in the publicly available FruitNeRF Dataset as testing examples (Fig. 2a and b). Details about the data creation can be found in FruitNeRF [4]. Since the dataset includes segmentation masks, camera intrinsics and poses, no additional processing was required.

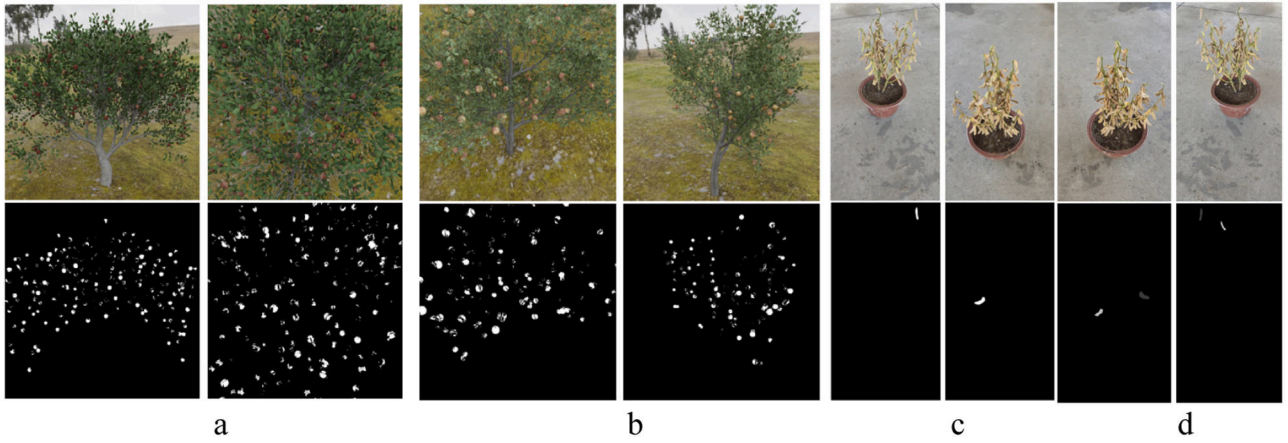
To further challenge the models, we collected a real-world soybean dataset specifically for this study. Video recordings of a soybean plant was captured using an iPhone 12 Pro under natural outdoor lighting conditions, with bright sunlight to reflect realistic field environments (Fig. 2c and d). The soybean plant was in a mature stage with clearly visible pods. The video was captured by manually moving a handheld camera around the soybean plant placed on the ground, ensuring multi-view coverage from different angles. Standard preprocessing was performed, including frame extraction from video using FFmpeg [16] at 5 frames per second and followed by camera pose estimation using COLMAP [17]. After discarding motion-blurred frames from the handheld capture, 149 clean RGB images of the plant were retained for reconstruction; camera poses were estimated with COLMAP, and this single reconstruction was used for both the single-pod and double-pod segmentation tasks, which share the same RGB images and differ only in their target masks (a single-pod mask or a double-pod mask). To generate pod segmentation masks, we used the AnyLabeling annotation tool with Segment Anything (MobileSAM) [18] integrated for automatic mask generation. Two tasks were defined using this dataset: (1) binary instance-level 3D segmentation of a single soybean pod, and (2) multi-instance 3D segmentation of two pods from the same plant. Two pods were selected as a controlled proof of concept for the multi-instance setting rather than as an exhaustive plant-level count; annotating every pod on the plant was not the objective of this feasibility validation and is constrained by severe pod-to-pod occlusion in a single handheld capture.

#### 4.2. Implementation details

##### 4.2.1. NeRF configuration

We adopted the Nerfacto model from Nerfstudio [19] using its default configuration settings (hidden\_dim = 64, hidden\_dim\_color = 64, appearance\_embed\_dim = 32). For the apple and peach datasets, all experiments were conducted using the default far-plane value of 1000 without any parameter tuning, demonstrating that InvNeRF-Seg operates effectively under standard NeRF configurations.

For the soybean plant dataset, where the target object occupies a relatively small spatial region compared to the full scene, the model capacity was increased (hidden dimensions raised to 128), and the far plane was reduced to 2 to restrict sampling to an object-centric volume. This adjustment is consistent with standard NeRF practice for sparse-object scenes, where excessive background sampling can dilute



**Fig. 2.** Examples of the synthetic data from FruitNeRF and real-world data of a soybean plant: apple (a), peach trees (b) and a soybean plant with a single pod (c) and two pods (d) annotated in different views (top row) and segmented fruit/pod masks (bottom row).

supervision signals and hinder effective optimization.

The RGB training stage was performed for 4000 epochs using learning rates of  $1 \times 10^{-2}$  for the field and proposal networks and  $1 \times 10^{-4}$  for the camera optimizer. The subsequent fine-tuning stage was conducted for an additional 6000 epochs, with all learning rates reduced by a factor of 100.

To ensure a fair comparison, all configuration settings—including model capacity, training schedule, and far-plane values—were kept identical between InvNeRF-Seg and FruitNeRF for each dataset. FruitNeRF was trained for a total of 6000 epochs to match the overall training budget.

#### 4.2.2. SA3D implementation

We implemented two variants of SA3D for comparison. The original weight-based SA3D back-projects 2D segmentation masks into 3D using volume-rendered weights with a threshold of  $\tau = 0.1$ , producing surface-like point clouds. In addition, we implemented a density-based variant that thresholds raw field density ( $\sigma \geq 70$ ) combined with 2D mask filtering, resulting in denser volumetric point clouds that better capture object interiors. This density-thresholded variant extracts points directly from the unmodified NeRF density field, followed by the same DBSCAN clustering pipeline used for all methods. It therefore serves as a geometry-based NeRF + DBSCAN baseline without any density-field refinement.

Both variants operate on the pretrained RGB NeRF model without any additional training or fine-tuning. The density-based variant is included to provide a more favorable setting for clustering-based downstream tasks, as volumetric representations are better suited for density-based clustering methods such as DBSCAN.

#### 4.2.3. Clustering and counting

All methods used an identical clustering pipeline to ensure fair comparison: DBSCAN clustering ( $\text{eps}=0.0065$ ,  $\text{min\_samples}=3$ ), followed by recursive splitting of large clusters using KMeans based on size and volume statistics. This ensures that counting accuracy differences reflect segmentation quality rather than post-processing choices.

#### 4.3. Evaluation metrics

Peak Signal-to-Noise Ratio (PSNR) [20] and Intersection over Union (IoU) [21] were used to evaluate the quality of the rendered 2D RGB images and masks, respectively, during training. The accuracy of fruit counting was used to evaluate the quality of generated 3D segmentation point clouds. A two-tailed t-test was performed using SciPy (v1.15.2) [22] to evaluate statistical significance.

#### 4.4. Experimental environment

All experiments were carried out on a Windows 11 Pro platform with Intel Core i9 5 GHz  $\times$  16 processor (CPU), 64 GB of RAM, and NVIDIA RTX A5500 GPU with 16 GB of RAM.

#### 4.5. Results on synthetic datasets

##### 4.5.1. Rendered image and mask quality

The quality of the rendered RGB images for both apple and peach datasets was comparable between InvNeRF-Seg and FruitNeRF, as indicated by the non-significant differences in PSNR values—despite slightly higher scores from InvNeRF-Seg (Fig. S1). This outcome is expected, as both models share the same architecture for RGB-based 3D reconstruction.

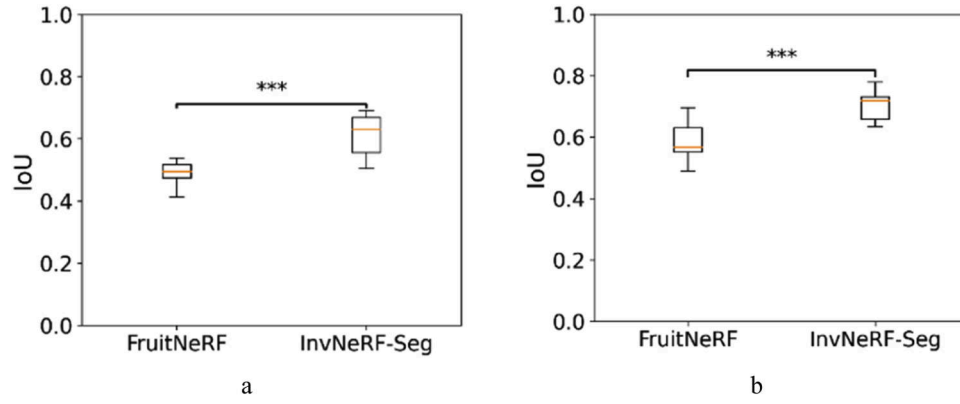
The IoU of the rendered masks for both apple and peach datasets was significantly higher in InvNeRF-Seg compared to FruitNeRF, with p values below 0.001—highlighting the superior segmentation accuracy of InvNeRF-Seg (Fig. 3). This is expected, as FruitNeRF remains dependent on RGB supervision, treating segmentation as a secondary task. In contrast, InvNeRF-Seg fine-tunes the density field directly using ground truth masks, allowing for more precise alignment between the learned radiance field and object boundaries.

##### 4.5.2. 3D point cloud quality

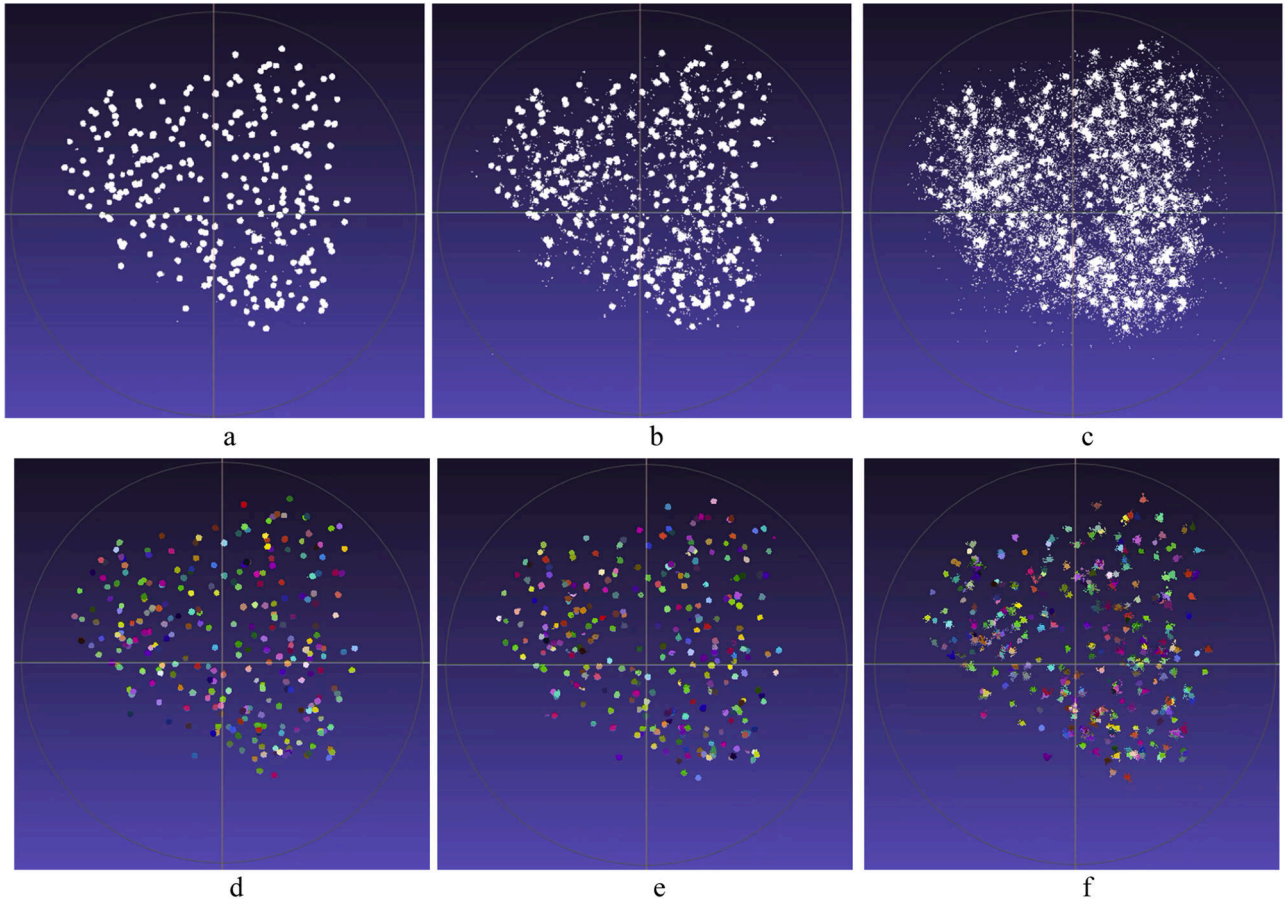
The clearest difference between methods is visible in the 3D segmented point clouds, apple (Fig. 4) and peach (Fig. S2). InvNeRF-Seg produces clean, well-separated fruit-shaped blobs with minimal noise. The instances in Fig. 4 are sufficiently well separated that a simple non-learned clustering method (DBSCAN) recovers near-exact fruit counts (282 vs. 283 for apple and 155 vs. 152 for peach) without requiring a learned detector. This demonstrates one practical advantage of reshaping the density field into discrete, well-separated object instances. FruitNeRF generates recognizable but fragmented fruits with scattered points around object boundaries. SA3D (both variants) produces heavily noisy point clouds with extensive scattered points throughout the scene. This quality difference is primarily attributable to the refined density field: InvNeRF-Seg's fine-tuning suppresses background density while reinforcing object density, whereas SA3D operates on an unmodified density field and FruitNeRF's joint training compromises density field quality.

##### 4.5.3. Object counting

Using the identical clustering pipeline across all methods, InvNeRF-Seg achieves the most accurate fruit counts (Table 1). The cleaner point clouds from InvNeRF-Seg naturally separate into individual fruit



**Fig. 3.** Comparison of IoU between FruitNeRF and InvNeRF-Seg for predicting masks of apple (a) and peach (b). ‘\*\*\*’ indicates  $p < 0.001$ .



**Fig. 4.** Comparison of segmented and clustered 3D point clouds for the apple dataset. Top row: segmented fruit point clouds (white) from InvNeRF-Seg (a), FruitNeRF (b), and SA3D (c). Bottom row: corresponding clustered point clouds with individual fruits shown in distinct colors from InvNeRF-Seg (d), FruitNeRF (e), and SA3D (f). Ground truth count: 283.

**Table 1**

Fruit counting results across methods using identical clustering parameters.

| Method          | Apple (GT=283) | Peach (GT=152) |
|-----------------|----------------|----------------|
| InvNeRF-Seg     | 282            | 155            |
| FruitNeRF (BCE) | 294            | 160            |
| FruitNeRF (MSE) | 289            | 156            |
| SA3D (density)  | 1573           | 1249           |
| SA3D (weight)   | 351            | 328            |

clusters, while noisier point clouds from other methods lead to

fragmentation and overcounting.

#### 4.6. Ablation studies

We conducted systematic ablation studies on the apple and peach datasets to understand why InvNeRF-Seg works and what each component contributes.

##### 4.6.1. Training from scratch on masks

Directly training NeRF on binary masks without pretrained RGB geometry produces completely degenerate outputs: entirely black

rendered images with no meaningful depth structure (Fig. 1). This confirms that pretrained geometry is essential and cannot be learned from binary masks alone. The IoU remains approximately zero throughout training.

#### 4.6.2. Component analysis: density vs. color MLP

Nerfacto's architecture consists of a density MLP (mlp\_base, 12.2 M parameters) and a color MLP (mlp\_head, 39 K parameters). We isolated the contributions of each by selectively freezing one during fine-tuning. The qualitative differences in 3D segmentation under different training strategies are illustrated in Fig. 5, where InvNeRF-Seg produces cleaner and more coherent point clouds compared to partial fine-tuning or joint training approaches.

**Fine-tune density only (color frozen).** Training progresses slowly, taking roughly 3000 epochs before IoU begins to improve. This is likely due to the large number of parameters in the density MLP, which makes optimization under mask-only supervision difficult. Nevertheless, modifying the density field is essential for forming clean and well-separated 3D point cloud structures.

**Fine-tune color only (density frozen).** Training converges quickly, reflecting the small size of the color MLP. However, since the geometry is fixed, the model produces spatially biased masks and flat depth maps. As a result, the extracted point clouds contain noticeably more noise.

**Fine-tuning both (InvNeRF-Seg).** Jointly updating both components leads to faster convergence and higher IoU, indicating that adapting color and density together is necessary for the best overall performance.

Counting results further validate this analysis (Table 2). While density-only fine-tuning achieves near-perfect counting on apple (281/

**Table 2**

Ablation study: fruit counting with different fine-tuning strategies.

| Fine-tuning Strategy | Apple (GT=283) | Peach (GT=152) |
|----------------------|----------------|----------------|
| Both (InvNeRF-Seg)   | 282            | 155            |
| Density only         | 281            | 157            |
| Color only           | 294            | 154            |

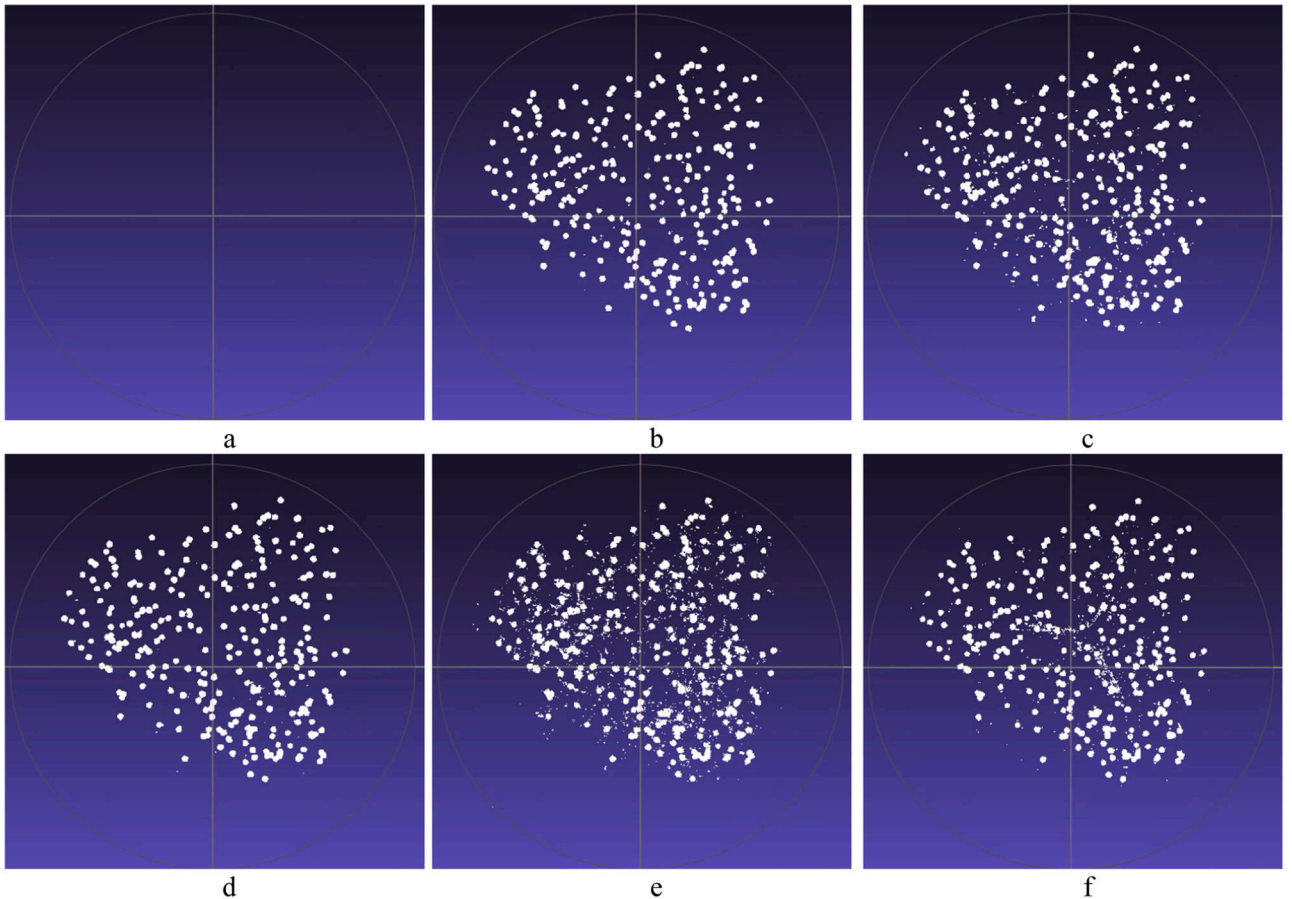
283), color-only fine-tuning significantly overcounts (294/283), indicating that density refinement is critical for producing well-separated 3D clusters suitable for downstream tasks.

#### 4.6.3. Loss function: MSE vs. BCE in joint training

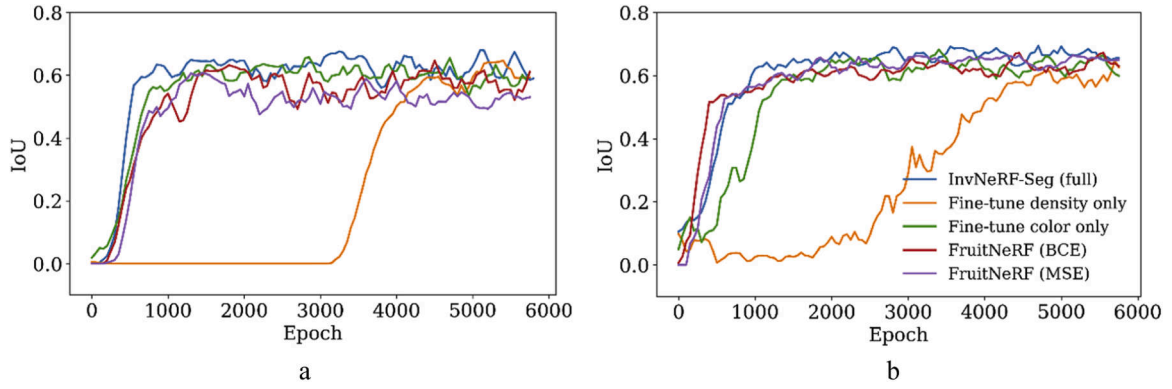
To isolate whether the advantage of InvNeRF-Seg comes from the two-stage strategy or the loss function, we trained FruitNeRF with MSE loss (replacing its default BCE) on the segmentation head. FruitNeRF with MSE achieved comparable final IoU to BCE but with slower initial convergence. Since both loss functions yield similar results under joint training, yet InvNeRF-Seg outperforms both, this demonstrates that the two-stage fine-tuning strategy—not the loss function choice—is the primary driver of InvNeRF-Seg's advantage.

#### 4.6.4. Convergence analysis

IoU convergence curves (Fig. 6) reveal distinct behaviors across all ablation conditions. InvNeRF-Seg (full fine-tuning) converges fastest and reaches the highest IoU. Fine-tune color only converges quickly but plateaus at a lower level. FruitNeRF (BCE and MSE) converges more slowly, lacking the pretrained geometry advantage for the fine-tuning stage. Fine-tune density only is dramatically slower, requiring approximately 3000 epochs before meaningful IoU improvement, but



**Fig. 5.** 3D segmented point clouds from the apple dataset under different training strategies: (a) train from scratch on masks (failed — no meaningful output), (b) fine-tune density only, (c) fine-tune color only, (d) InvNeRF-Seg (full fine-tuning), (e) FruitNeRF (BCE), (f) FruitNeRF (MSE).



**Fig. 6.** IoU convergence curves during training for all ablation conditions on apple (a) and peach (b): InvNeRF-Seg (full), fine-tune density only, fine-tune color only, FruitNeRF (BCE), and FruitNeRF (MSE).

eventually reaches competitive final values. These results confirm that pretrained geometry provides a critical head start, and that co-adapting both MLPs yields the best convergence profile.

#### 4.7. SA3D variant comparison

We compared the original weight-based SA3D with our density-based variant. The density-based variant produces cleaner, denser point clouds because raw density captures object interiors rather than surfaces only. Despite this more favorable formulation for SA3D, InvNeRF-Seg still substantially outperforms both SA3D variants in downstream counting (Table 1), confirming that the advantage comes from density field refinement through fine-tuning, not from the point cloud extraction strategy.

#### 4.8. Real-World validation: soybean dataset

To evaluate robustness under challenging conditions, we tested InvNeRF-Seg on a self-collected real-world soybean dataset acquired under natural conditions for binary instance-level 3D segmentation of a single pod. InvNeRF-Seg achieved comparable performance in rendering RGB images (Fig. 7a), while producing substantially higher-quality segmentation masks during training compared to FruitNeRF ( $p < 0.01$ ) (Fig. 7b). Single-pod segmentation quality is further quantified by rendered-mask IoU and PSNR during training (Fig. 7), where InvNeRF-Seg significantly exceeds FruitNeRF in mask IoU ( $p < 0.01$ ). The segmented 3D single pod generated by InvNeRF-Seg also appeared visibly cleaner and less noisy than those produced by both FruitNeRF and SA3D, based on visual inspection (Fig. 8 and Fig. S3). These results further confirm the robustness and practical applicability of the

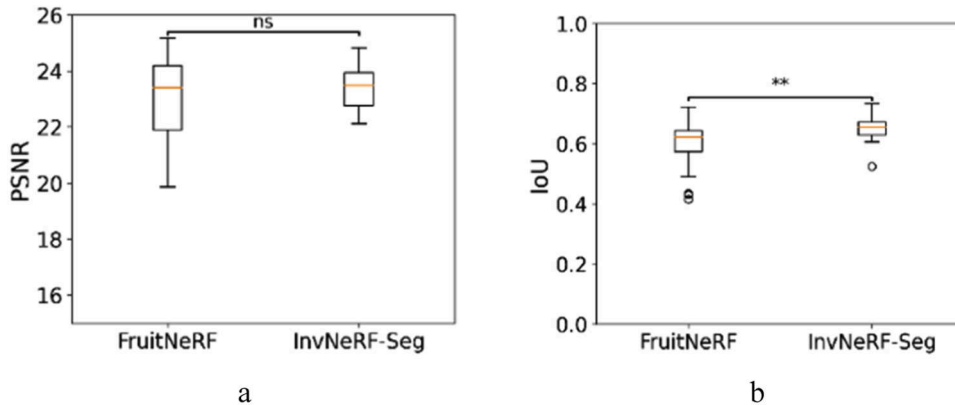
proposed approach in real-world agricultural scenarios. Because no dense 3D ground truth is available for the field-collected plant, this 3D point cloud is assessed qualitatively.

#### 4.9. Extension to multi-instance segmentation

Although InvNeRF-Seg was originally designed for binary segmentation, we further tested its ability to handle multi-instance segmentation by separating two distinct pods from the same soybean plant. Despite using the same MSE loss and unchanged model architecture, the method produced promising results for double-pod segmentation. However, its performance was slightly lower than in the binary case, as evidenced by some misclassified points near the instance boundaries (Fig. 9). SA3D also performed reasonably well in this task, although its outputs exhibited noticeable noise across both object and background regions.

#### 4.10. Field Density Change Analysis

The volumetric field density before and after fine-tuning was analyzed to better understand the internal behavior of InvNeRF-Seg. Targeted object regions consistently show increased density after fine-tuning, leading to sharper and more localized peaks along sampled rays (Fig. 10a-f). In contrast, non-targeted objects and background areas exhibit field density suppression or small fluctuations around zero, which results in minimal accumulated weights and effective background removal (Fig. 10g-l). These patterns confirm that InvNeRF-Seg semantically reshapes the radiance field without requiring architectural changes.



**Fig. 7.** Comparison of PSNR (a) and IoU (b) for single pod segmentation from a soybean plant between FruitNeRF and InvNeRF-Seg. 'ns' indicates no statistical significance; '\*\*' indicates  $p < 0.01$ .

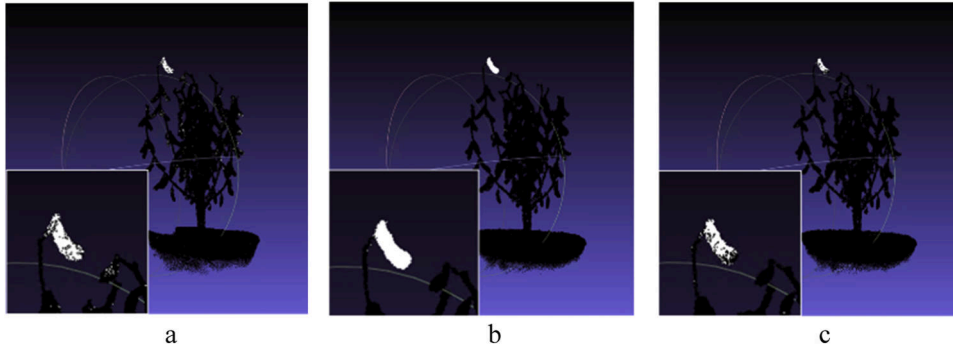


Fig. 8. Segmented single-pod 3D point clouds from a soybean plant obtained with InvNeRF-Seg (a), FruitNeRF (b) and SA3D (c).

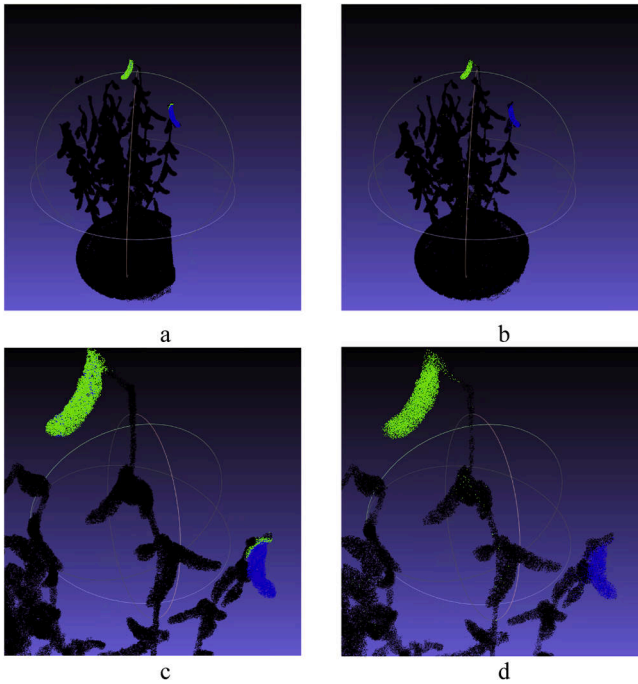


Fig. 9. Segmented double-pod 3D point clouds from a soybean plant obtained with InvNeRF-Seg (a, c) and SA3D (b, d); the two pods are separated by instance.

## 5. Discussion

Our experiments across both synthetic and real-world datasets demonstrate that InvNeRF-Seg, a two-step fine-tuning strategy that leverages a pre-trained RGB NeRF, consistently outperforms existing 3D segmentation methods—namely SA3D [1] and the joint training-based FruitNeRF [4]—in terms of segmentation accuracy, 3D point cloud quality, and object-level counting performance.

**Decoupling geometry and semantics.** The fundamental insight is that geometry learning and semantic learning should be decoupled. NeRF models rely on smooth color gradients in RGB images to learn scene geometry; binary segmentation masks with hard transitions provide limited geometric cues and can hinder training if used too early. Joint learning, as in FruitNeRF, forces the model to optimize for conflicting objectives—RGB fidelity and semantic separation—using shared features, which can lead to suboptimal performance on both tasks [23]. In contrast, InvNeRF-Seg preserves geometric integrity by first optimizing for RGB reconstruction, and only then introduces semantic supervision during fine-tuning.

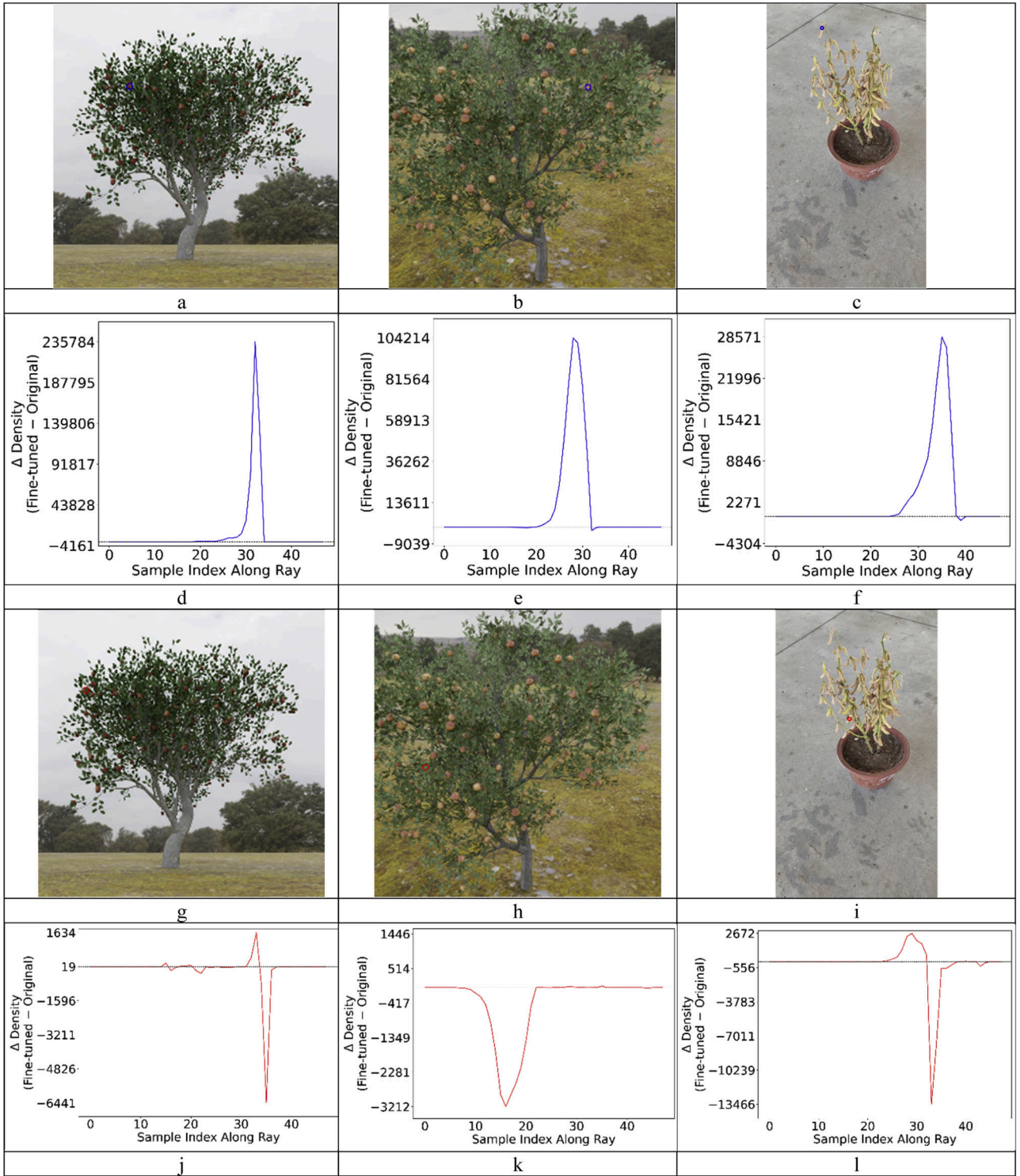
**Nature of the contribution.** InvNeRF-Seg is not a new network

architecture but a training strategy that exploits an empirical property of converged NeRF models: once a smooth density field has been learned from RGB images, it can be reshaped toward object regions by continued optimization of the unchanged MSE objective on mask-substituted inputs. The contribution is therefore twofold—an empirical finding about the adaptability of pretrained radiance fields, and a simple, architecture-invariant recipe that turns this finding into a practical 3D segmentation tool. We make no claim of a new loss or module; the value lies in showing that none is needed.

**Why input substitution reshapes the density field.** The mechanism follows from the volume-rendering objective. Under MSE supervision, optimization encourages the accumulated rendering contribution to match the binary target masks. Through the volume-rendering formulation, this results in increased density in foreground regions and reduced density in background regions, consistent with the density redistribution observed in Fig. 10. Starting from a pretrained field is essential: the gradients sharpen an existing surface rather than having to discover geometry from a near-binary signal, which is why training from scratch collapses (Fig. 1). This view also predicts when substitution fails—when foreground and background project onto the same rays (thin or heavily occluded structures), when 2D masks are inconsistent across views (the field cannot satisfy contradictory targets and blurs the boundary), and when the sampling volume greatly exceeds the object, so that most samples receive an uninformative background target and the effective foreground gradient is diluted, the same effect addressed by the far-plane setting discussed below.

**Why density adapts rather than colour.** A natural question is why, under the unchanged MSE objective, the network satisfies the mask target by reshaping the density field rather than by simply having the colour MLP emit white on foreground samples and black elsewhere. Both pathways receive gradients, but they are not equally effective. The colour MLP can only set the emitted value of a sample that is already being rendered; it cannot remove the contribution of background or occluding matter along the ray, because that contribution is governed by density and transmittance rather than colour. To render a clean binary image consistently across all views, the network must therefore concentrate accumulated weight on the object surface and drive it to zero elsewhere—an operation only the density field can perform, since a purely colour-based solution would have to encode a consistent foreground/background label for every 3D sample under every viewing direction, which the geometry represents far more parsimoniously. The ablations confirm this: colour-only fine-tuning leaves the density field diffuse and overcounts (294 versus a ground truth of 283), whereas density adaptation yields the clean, separable clusters required for accurate counting (Fig. 5; Table 2).

**Both MLPs contribute complementary roles.** Our ablation studies reveal that both density and color MLPs play complementary roles during fine-tuning. The density MLP reshapes the volumetric structure, which is critical for clean 3D point cloud extraction—as evidenced by density-only fine-tuning achieving near-perfect counting (281/283



**Fig. 10.** Field density changes after fine-tuning in targeted (object) and non-targeted (background) regions. Top two rows: example input pixels from apple, peach, and soybean pod, including targeted object regions (a–c) and background/non-targeted regions (g–i). Bottom two rows: corresponding per-ray field density differences for object regions (d–f) and background regions (j–l). Positive values indicate density enhancement in targeted object regions, whereas negative values indicate density suppression in background and non-targeted regions.

apples) despite slow convergence. The color MLP quickly adapts the rendered output to match binary targets. Full fine-tuning achieves the best results because both adaptations reinforce each other. Density-only fine-tuning is therefore a viable, simpler alternative when counting is the sole objective and a longer training budget is acceptable, since it attains comparable counting accuracy; full fine-tuning is preferred when both faster convergence and higher rendered-mask IoU are required.

**Point cloud quality matters >2D IoU.** While convergence curves show that several methods achieve similar final 2D IoU values, the 3D point cloud quality differs dramatically. InvNeRF-Seg produces clean, well-separated clusters that naturally correspond to individual objects, whereas FruitNeRF and SA3D produce fragmented or noisy point clouds that challenge downstream clustering. This highlights that 2D metrics alone are insufficient for evaluating NeRF-based 3D segmentation

methods.

**Volume control is critical.** Setting an appropriately small far plane is essential for stable fine-tuning, especially when the target object occupies a small portion of the scene. For instance, fine-tuning for single-pod segmentation consistently failed when using a large far plane (e.g., 1000), but succeeded reliably when the far plane was reduced to 2. The far plane must be consistent between RGB NeRF training and fine-tuning. The reason is that a large far plane places most sampled points in empty background whose mask target is zero; the background term then dominates the loss and the sparse foreground gradient cannot build a coherent density peak, which is why single-pod fine-tuning failed at a far plane of 1000 but succeeded at 2. In practice the far plane need not be found by trial and error: it can be set directly from the reconstruction geometry—for example from the extent of the COLMAP sparse point cloud or the depth range rendered by the Stage-1 model—choosing a value that just encloses the object of interest. The apple and peach scenes required no adjustment because the fruit-bearing tree fills the reconstructed volume, so the default far plane already concentrates sampling on foreground.

**Relation to direct 3D reconstruction.** InvNeRF-Seg does depend on 2D masks, but this is a mild requirement in practice: 2D segmentation is mature and can be produced automatically with foundation models such as SAM, whereas direct 3D segmentation would require annotated point clouds—the very bottleneck this work aims to avoid—or a separately reconstructed and registered 3D model that still has to be segmented. By lifting inexpensive 2D masks into a geometry-consistent 3D field, the method converts a well-supported 2D task into 3D labels with no manual 3D annotation. The trade-off is that segmentation quality is upper-bounded by the multi-view consistency of the input masks and by the fidelity of the underlying reconstruction.

**Choice of baselines.** SA3D and FruitNeRF were selected because they represent the two established paradigms for NeRF-based 3D segmentation—training-free post-processing of a fixed field (SA3D) and joint RGB-plus-semantics training (FruitNeRF)—and together bracket the design space that InvNeRF-Seg occupies. We additionally evaluated a density-thresholded SA3D variant (raw NeRF density with 2D-mask filtering, clustered with DBSCAN) as a stronger volumetric baseline. Feature-field distillation methods that lift 2D foundation-model features into NeRF are a promising further comparison, but they target open-vocabulary querying rather than the single-object, count-oriented setting studied here; a broader benchmark against them is left to future work.

**Limitations.** InvNeRF-Seg assumes access to accurate multi-view RGB images and corresponding 2D segmentation masks. The multi-instance extension using intensity-encoded masks shows boundary ambiguity due to the regression nature of MSE [24]. Incorporating class-aware losses such as cross-entropy may improve multi-instance performance, although this would break the “zero-change” design. The method has been validated primarily on agricultural datasets; evaluation on general-purpose scenes with more diverse object categories remains future work. Our soybean masks were generated with MobileSAM and manually verified. More broadly, the proposed two-stage segmentation is, by design, dependent on two high-quality inputs—a high-fidelity RGB NeRF reconstruction and accurate 2D segmentation masks—both of which are standard, well-supported components obtainable with current NeRF pipelines and foundation-model segmenters, and are treated here as prerequisites of the method rather than variables it must be robust to. Characterizing the precise input-quality threshold below which the pipeline fails is not well defined, since neither reconstruction nor segmentation quality reduces to a single tunable axis and the tolerance depends on scene geometry, object scale, occlusion, and view distribution; we therefore scope the method to the good-quality-input regime and make no claim outside it. Point-cloud quality is reported here through downstream counting accuracy, as dense 3D metrics such as Chamfer distance or 3D IoU require per-point ground truth, which neither the FruitNeRF synthetic data (annotated by

fruit count) nor the field-collected soybean data provide. Finally, the real-world validation is deliberately small in scale—one plant and one to two pods—and is intended to demonstrate feasibility under field conditions rather than large-scale performance; because no dense 3D ground truth exists for the field plant, its reconstructions are assessed by rendered-mask IoU/PSNR during training and by qualitative 3D inspection, and evaluation on multi-plant, densely-podded canopies remains future work.

## 6. Conclusions

We proposed InvNeRF-Seg, a two-step, zero-change fine-tuning strategy for NeRF-based 3D object segmentation that preserves the original network architecture and loss function. Through systematic ablation studies, we demonstrated that: (1) pretrained geometry is essential—training on masks from scratch fails completely; (2) both density and color MLPs must co-adapt during fine-tuning for optimal performance; and (3) the two-stage approach outperforms joint training regardless of loss function choice (MSE or BCE).

InvNeRF-Seg consistently produced cleaner 3D segmented point clouds than the compared methods, enabling accurate downstream object counting with minimal post-processing tuning. Field-level analysis reveals reinforced density in target regions and suppression of background content, confirming effective semantic refinement within the radiance field. Moreover, the method extends naturally from binary instance-level to multi-instance segmentation tasks, demonstrating generalizability across scenes of varying complexities.

This zero-change, modular strategy provides a practical architecture-invariant framework for bridging 2D segmentation and 3D semantic reconstruction.

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## CRedit authorship contribution statement

**Jiangsan Zhao:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Jakob Geipel:** Funding acquisition, Writing – review & editing. **Krzysztof Kusnierek:** Funding acquisition, Project administration, Writing – review & editing. **Xuean Cui:** Conceptualization, Funding acquisition, Project administration, Writing – review & editing.

## Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jiangsan Zhao reports financial support was provided by Research Council of Norway. Xuean Cui reports financial support was provided by Innovation Program of Chinese Academy of Agricultural Sciences. Xuean Cui has patent pending to Chinese Academy of Agricultural Sciences. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Supplementary materials

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## Data availability

Data will be made available on request.

## References

- [1] J. Cen, J. Fang, Z. Zhou, C. Yang, L. Xie, X. Zhang, W. Shen, Q. Tian, Segment anything in 3d with radiance fields, *Int. J. Comput. Vis.* 133 (2025) 5138–5160.
- [2] B. Mildenhall, P.P. Srinivasan, M. Tancik, J.T. Barron, R. Ramamoorthi, R. Ng, Nerf: representing scenes as neural radiance fields for view synthesis, *Commun. ACM* 65 (2022) 99–106.
- [3] A. Pumarola, E. Corona, G. Pons-Moll, F. Moreno-Noguer, -nerf: neural radiance fields for dynamic scenes, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 10318–10327.
- [4] L. Meyer, A. Gilson, U. Schmid, M. Stamminger, Fruitnerf: a unified neural radiance field based fruit counting framework, in: *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2024, pp. 1–8.
- [5] S. Zhi, T. Laidlow, S. Leutenegger, A.J. Davison, In-place scene labelling and understanding with implicit scene representation, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 15838–15847.
- [6] Y. Jiang, G. Liu, Z. Huang, B. Yang, W. Yang, Geometry perception and motion planning in robotic assembly based on semantic segmentation and point clouds reconstruction, *Eng. Appl. Artif. Intell.* 130 (2024) 107678.
- [7] Y. Wen, A.B. Shah, R. Cao, C. Zhang, J. Chen, X. Wu, C. Xie, X. Fu, AR-Light: enabling fast and lightweight multi-user augmented reality via semantic segmentation and collaborative view synchronization, *IEEE Trans. Comput.* 74 (2025) 2073–2086.
- [8] Y. Zhuang, Q. Li, Y. Chen, J. Huai, M. Li, T. Ma, Y. Tang, X. Liang, 3D-SeqMOS: a novel sequential 3D moving object segmentation in autonomous driving, *IEEE Trans. Intell. Transp. Syst.* 25 (2024) 8782–8795.
- [9] W. Li, H. Xu, G. Zhang, H. Gao, M. Gao, M. Wang, H. Zhao, Fairdiff: fair segmentation with point-image diffusion, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, 2024, pp. 617–628.
- [10] R. Yang, Y. He, X. Lu, Y. Zhao, Y. Li, Y. Yang, W. Kong, F. Liu, 3D-based precise evaluation pipeline for maize ear rot using multi-view stereo reconstruction and point cloud semantic segmentation, *Comput. Electron. Agric.* 216 (2024) 108512.
- [11] C.R. Qi, H. Su, K. Mo, L.J. Guibas, Pointnet: deep learning on point sets for 3d classification and segmentation, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 652–660.
- [12] C.R. Qi, L. Yi, H. Su, L.J. Guibas, Pointnet++: deep hierarchical feature learning on point sets in a metric space, in: *Advances in Neural Information Processing Systems* 30, 2017.
- [13] T. Miao, W. Wen, Y. Li, S. Wu, C. Zhu, X. Guo, Label3Dmaize: toolkit for 3D point cloud data annotation of maize shoots, *Gigascience* 10 (2021) giab031.
- [14] C. Liu, D. Zeng, A. Akbar, H. Wu, S. Jia, Z. Xu, H. Yue, Context-aware network for semantic segmentation toward large-scale point clouds in urban environments, *IEEE Trans. Geosci. Remote Sens.* 60 (2022) 1–15.
- [15] L. Pan, ECG: edge-aware point cloud completion with graph convolution, *IEEE Robot. Autom. Lett.* 5 (2020) 4392–4398.
- [16] S. Tomar, Converting video formats with Ffmpeg, *Linux J.* 2006 (2006) 10.
- [17] J.L. Schönberger, J.-M. Frahm, Structure-from-motion revisited, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 4104–4113.
- [18] Zhang, C., Han, D., Qiao, Y., Kim, J.U., Bae, S.-H., Lee, S., Hong, C.S., 2023. Faster segment anything: towards lightweight sam for mobile applications. *arXiv Prepr. arXiv2306.14289*.
- [19] M. Tancik, E. Weber, E. Ng, R. Li, B. Yi, T. Wang, A. Kristoffersen, J. Austin, K. Salahi, A. Ahuja, Nerfstudio: a modular framework for neural radiance field development, in: *ACM SIGGRAPH 2023 Conference Proceedings*, 2023, pp. 1–12.
- [20] J. Korhonen, J. You, Peak signal-to-noise ratio revisited: is simple beautiful?, in: *2012 Fourth International Workshop on Quality of Multimedia Experience*, 2012, pp. 37–38. IEEE.
- [21] R. Girshick, Fast r-cnn, in: *Proceedings of the IEEE International Conference on Computer Vision*, 2015, pp. 1440–1448.
- [22] P. Virtanen, R. Gommers, T.E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, SciPy 1.0: fundamental algorithms for scientific computing in Python, *Nat. Methods.* 17 (2020) 261–272.
- [23] T. Standley, A. Zamir, D. Chen, L. Guibas, J. Malik, S. Savarese, Which tasks should be learned together in multi-task learning?, in: *International Conference on Machine Learning*. PMLR, 2020, pp. 9120–9132.
- [24] J. Zhao, A. Kumar, B.N. Banoth, B. Marathi, P. Rajalakshmi, B. Rewald, S. Ninomiya, W. Guo, Deep-learning-based multispectral image reconstruction from single natural color RGB image—enhancing UAV-based phenotyping, *Remote Sens.* 14 (2022) 1272.